T. H. MIHIRAN THILAKARATHNA

IT Undergraduate | AI/ML & Full-Stack Developer
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PROFESSIONAL SUMMARY

Results-driven IT undergraduate at Rajarata University of Sri Lanka with hands-on experience in AI, Machine Learning, and full-stack software development. Proficient in end-to-end ML pipelines (TensorFlow, PyTorch, Scikit-learn) with expertise in Computer Vision, NLP, and on-device inference via TensorFlow Lite. Builds production-grade applications across Android (Kotlin/Jetpack Compose), web (React, Node.js, MongoDB), and ML model conversion platforms. Grand Finalist at InnovateX 2025 (Zebra Technologies). Seeking an AI/ML internship to apply expertise in real-world intelligent systems.

EDUCATION

Rajarata University of Sri Lanka, Mihintale 2023 – Present
BSc. Information Technology | 3rd Year Undergraduate

WORK EXPERIENCE

Banking Operations Trainee Dec 2022 – Jul 2023
People's Bank, Avissawella Branch

- Assisted with daily banking operations, financial documentation, and customer service in a regulated financial environment.
- Strengthened professional communication, teamwork, and attention-to-detail skills applicable to enterprise software environments.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++, SQL
AI / ML: Machine Learning, Deep Learning, NLP, Computer Vision, LLM APIs, Data Preprocessing, Model Training & Evaluation
Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, Hugging Face
Web & Mobile: React, Node.js, Express.js, Jetpack Compose (Kotlin), Flutter
Databases: MySQL, MongoDB, SQLite, Firebase (Cloud Firestore)
Tools: Git, Jupyter Notebook, Google Colab, VS Code, Vercel, Render, Hugging Face Spaces

PROJECTS

OmniModel Converter GitHub

- Full-stack ML model conversion platform automating translation across PyTorch, TensorFlow, ONNX, and TFLite via FastAPI backend (Hugging Face Spaces) and Next.js frontend (Vercel).
- **Tech:** Python, FastAPI, PyTorch, TensorFlow, ONNX, TFLite, Next.js

Lumo AI Assistant GitHub

- AI-powered Android assistant (Jetpack Compose / Kotlin) integrating Google Gemini API for intelligent conversations, PDF processing, Firebase Auth, and Cloud Firestore.
- **Tech:** Kotlin, Jetpack Compose, Firebase, Google Gemini API, PDFBox-Android

MoodTune — Emotion-Aware Music Recommender GitHub

- Android app using a custom CNN (FER-2013 dataset) via TensorFlow Lite and Google ML Kit for real-time facial emotion detection with mood-matched music recommendations.
- **Tech:** Java, TensorFlow Lite, Google ML Kit, Python/Keras, CameraX, MVVM

Facial Occlusion Classification for Restricted Environments GitHub

- MobileNetV2 transfer learning model achieving 98% test accuracy classifying faces as clear, helmet-occluded, or mask-occluded — deployed with a Tkinter GUI for real-time inference.
- **Tech:** Python, TensorFlow, Keras, MobileNetV2, Scikit-learn, PIL, Matplotlib

Vendora — SaaS E-Commerce Admin Dashboard GitHub

- Full-stack admin dashboard with RBAC, product/order management, and real-time notifications; frontend on Vercel, backend on Render.
- **Tech:** React 18, Node.js, Express.js, MongoDB, Tailwind CSS

ACHIEVEMENTS & AWARDS

Grand Finalist — InnovateX 2025 2025

- Selected as a Grand Finalist at InnovateX 2025 (Zebra Technologies, Galle Face Hotel) — one of Sri Lanka's premier technology innovation competitions, representing Rajarata University as Team LoopCode.